MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 9

1. *For each of the following matrices, find the characteristic polynomial, eigenvalues and corresponding eigenvectors (i.e., for each eigenvalue give a maximal set of linearly independent eigenvectors which correspond to that eigenvalue):*

(1) $\begin{bmatrix} 3 & -1 \\ 1 & 5 \end{bmatrix}$

(2) $\begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

(3) $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

(4) $\begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix}$

2. *True/False. Give reasons why! Below A and B are matrices of an appropriate size.*

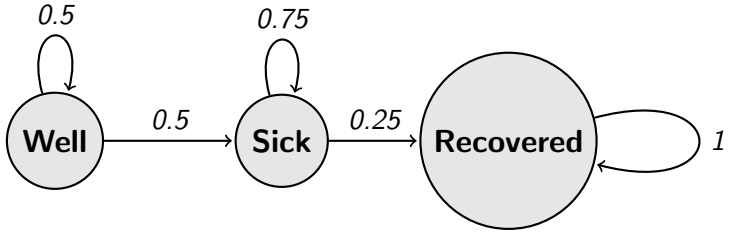
1. *A square matrix A is not invertible iff 0 is an eigenvalue of A.*
2. *The eigenvalues of a square matrix are the numbers on its main diagonal.*
3. *If $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent eigenvectors of a square matrix A, then they correspond to different eigenvalues.*
4. *The sum of two eigenvectors of a matrix A is also an eigenvector of A.*

3. *Let A, B be two $n \times n$-matrices such that $AB = BA$.*

(a) Suppose $\boldsymbol{v}$ is an eigenvector of A with eigenvalue λ. Is $B\boldsymbol{v}$ an eigenvector of A? If so, what is the eigenvalue of that eigenvector?

(b) Suppose A has eigenvectors $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ with distinct eigenvalues $\lambda_1 \neq \ldots \neq \lambda_n$. Is each $\boldsymbol{v_i}$ also an eigenvector of B? (This question is a bit tricker. Hint: Note that each of the eigenspaces of A has dimension 1 and then use your answer to (a).).

4. *Suppose there is an epidemic with the following transition graph:*



Set up a 3×3 transition matrix A and find the steady state $\boldsymbol{v}_\infty$.